

Practice Questions from Week 1

Question 2 : 640653738030

[View Solutions \(0\)](#) Total Mark : 4.00 | Type : MCQ

Suppose a dataset lies in $\mathbb{R}^4$ and undergoes Principal Component Analysis (PCA) after being centered. The resulting first and second principal components are given by:

$$\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

Which of the following could be the third principal component?

OPTIONS :

☐ $\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ [ⓐ](#)

☐ $\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ [ⓐ](#)

☐ $\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$ [ⓐ](#)

☐ $\frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ [ⓐ](#)

Solution(2)

we know, all principal components are orthogonal to each other i.e, $w_i^T w_j = 0 \quad \forall i \neq j$

$$w_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}; w_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$w_1^T w_2 = 0$$

$$w_1^T b = \frac{1}{\sqrt{3}} [1 \quad -1 \quad 0 \quad 1] \times \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \left[\frac{1}{3} - \frac{1}{3} + 0 + 0 \right] = 0$$

$$w_2^T b = \frac{1}{\sqrt{3}} [1 \quad 0 \quad 1 \quad -1] \times \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \left[\frac{1}{3} + 0 + \frac{1}{3} + 0 \right] = \frac{2}{3}$$

similarly workout for option (a), (c), (d)

$$w_1^T w_2 = w_1^T d = w_2^T d = 0$$

Question 6 : 640653738031

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Standard PCA has been performed on a centered dataset in $\mathbb{R}^3$. The first two principal components are given below:

$$\mathbf{w}_1 = \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{w}_2 = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Consider the data point in the dataset: $[2 \ 1 \ -1]^T$. (a, b) is the representation of this point in the coordinate system formed by the two principal components given above. The first and second coordinates correspond to PC-1 and PC-2 respectively.

[Discussions \(0\)](#)**Question 7 : 640653738032**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

What is the value of a? Enter your answer correctly to three decimal places.

Answer (Numeric):

[Discussions \(0\)](#)**Question 8 : 640653738033**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

What is the value of b? Enter your answer correctly to two decimal places.

Answer (Numeric):

Solution(7 and 8)

Given,

$$w_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} ; w_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} ; x = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow a = w_1^T x = \frac{1}{\sqrt{3}} [1 \ 1 \ 1] \times \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \left[\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} \right] = 1.15$$

$$\Rightarrow b = w_2^T x = \frac{1}{\sqrt{2}} [1 \ -1 \ 0] \times \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \left[\frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 0 \right] = 0.70$$

Question 2 : 640653827433

[View Solutions \(0\)](#) Total Mark : 2.00 | Type : MCQ

Let $\mathbf{w}_1$ and $\mathbf{w}_2$ be the top two principal components of the covariance matrix of a centered dataset. Let $\mathbf{x} \in \mathbb{R}^d$ be a data-point. Which of the following is the reconstruction error of the data-point after projecting it onto the top two principal components?

OPTIONS :

☐ $\|\mathbf{x} - (\mathbf{x}^T \mathbf{w}_1) \mathbf{w}_1 - (\mathbf{x}^T \mathbf{w}_2) \mathbf{w}_2\|^2$

☐ $\|\mathbf{x} - (\mathbf{x}^T \mathbf{w}_1) \mathbf{w}_1\|^2 + \|\mathbf{x} - (\mathbf{x}^T \mathbf{w}_2) \mathbf{w}_2\|^2$

☐ $(\mathbf{x}^T \mathbf{w}_1)^2 + (\mathbf{x}^T \mathbf{w}_2)^2$

☐ $\|\mathbf{x} - \mathbf{w}_1\|^2 + \|\mathbf{x} - \mathbf{w}_2\|^2$

Solution(2)

projection of x into w_i principal component = $(x^T w_i) w_i$

The reconstructed data point using the top two principal components is

$$\implies x' = (x^T w_1) w_1 + (x^T w_2) w_2$$

reconstruction error formulae

$$\implies \|x - x'\|^2$$

$$\implies \|x - (x^T w_1) w_1 - (x^T w_2) w_2\|^2$$

Question 4 : 640653827435

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : MSQ

Consider a covariance matrix C for a mean-centered dataset in $\mathbb{R}^3$. After performing

standard PCA, the three principal components turn out to be $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Which of the following statements are true? You can assume that C is not the zero matrix.

OPTIONS :

☐ C is a diagonal matrix.

☐ The diagonal entries of C are non-negative.

☐ The diagonal entries of C are strictly greater than zero.

☐ C has to be the identity matrix.

☐ C is a matrix of the form kI , where $k > 0$ and I is the identity matrix.

Solution(4)

Option 1 : C is a diagonal matrix

True, The eigenvectors are aligned with the coordinate axes, indicating C is diagonal.

Option 2 : The diagonal entries of C are non – negative

True, Covariance matrices are positive semi – definite, ensuring all eigenvalues (diagonal entries)

are non – negative.

Option 3 : The diagonal entries of C are strictly greater than zero

False, While $C \neq 0$, it is possible for some eigenvalues to be 0

Option 4 : C has to be the identity matrix

False, C being diagonal does not imply it must be I; the eigenvalues can vary.

Option 5 : C is of the form kI , where $k > 0$

False, The eigenvalues may not all be equal.

Question 9 : 640653827440

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let C be the covariance matrix of a mean-centered dataset: 

$$C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Standard PCA is performed on this dataset. The first two

PCs are given below:

$$w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad w_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

[Discussions \(0\)](#)**Question 10 : 640653827441**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Find the variance of the dataset along the first principal component.

Answer (Numeric):

[Discussions \(0\)](#)**Question 11 : 640653827442**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Find the variance of the dataset along the x-axis.

Answer (Numeric):

Solution (10)

We know, $\lambda_1 = w_1^T C w_1$

$$\Rightarrow \lambda_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3$$

Solution (11)

The *variance-covariance* matrix has the following structure:

$$\begin{bmatrix} \text{var}(x) & \text{cov}(x, y) \\ \text{cov}(x, y) & \text{var}(y) \end{bmatrix}$$

$$\Rightarrow C_{11} = 2$$

Q. Any alternative Way ?

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow x^T C x \Rightarrow 2$$

Q. What will be the variance of the dataset along y-axis?

$$y \Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow y^T C y \Rightarrow 2$$

Question 12 : 640653827443

 [View Parent QN](#)

 [View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Find the coordinates of the point $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ in the new coordinate system formed by the principal components.

OPTIONS :

☐ $\begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

☐ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

☐ $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

Solution(12)

$$1^{st} \text{ coordinate} = (x^T w_1) = \left([2 \ 2] \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$2^{nd} \text{ coordinate} = (x^T w_2) = \left([2 \ 2] \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right) = 0$$

Question 5 : 640653992016

[View Solutions \(0\)](#) Total Mark : 4.00 | Type : MSQ

Which of the following expressions is the reconstruction error for a dataset of n points, with respect to a line passing through the origin represented by the vector $\mathbf{w}$.
Note that $\|\mathbf{w}\| = 1$.

OPTIONS :

☐ $\frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - (\mathbf{x}_i^T \mathbf{w}) \mathbf{w}\|^2$

☐ $\frac{1}{n} \sum_{i=1}^n [\mathbf{x}_i - (\mathbf{x}_i^T \mathbf{w}) \mathbf{w}]^T [\mathbf{x}_i - (\mathbf{x}_i^T \mathbf{w}) \mathbf{w}]$

☐ $\frac{1}{n} \sum_{i=1}^n [\mathbf{x}_i^T \mathbf{x}_i - (\mathbf{x}_i^T \mathbf{w})^2]$

☐ $-\frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w})^2$

Solution(5)

$$\begin{aligned} \text{Error}(\text{line}, \text{dataset}) &= \sum_{i=1}^n \text{error}(\text{line}, x_i) \\ &= \sum_{i=1}^n \text{length}^2(x - (x^T w)w) \\ &= \sum_{i=1}^n \|x_i - (x_i^T w)w\|^2 \\ &= \frac{1}{n} \sum_{i=1}^n \|x_i - (x_i^T w)w\|^2 \\ &= \frac{1}{n} \sum_{i=1}^n (x_i - (x_i^T w)w)^T (x_i - (x_i^T w)w) \\ &= \frac{1}{n} \sum_{i=1}^n [x_i^T x_i - (x_i^T w)^2 - (x_i^T w)^2 + (x_i^T w)^2 w^T w] \\ &= \frac{1}{n} \sum_{i=1}^n x_i^T x_i - (x_i^T w)^2 \end{aligned}$$

Question 7 : 640653992017

 **View Solutions (0)** Total Mark : 3.00 | Type : SA

The eigenvalues of the covariance matrix of a centered dataset in $\mathbb{R}^5$ are 30, 10, 10, 0, 0. Standard PCA is performed on this dataset. What is the variance captured by the top two principal components expressed as a percentage of total variance?



Answer (Numeric):

Answer

Solution(7)

$$\Rightarrow \frac{\lambda_1 + \dots + \lambda_k}{\lambda_1 + \dots + \lambda_d} = \frac{30 + 10}{30 + 10 + 10 + 0 + 0} = \frac{40}{50} = 0.8$$

$$\Rightarrow 0.8 \times 100 = 80$$

Question 11 : 640653992013

Total Mark : 0.00 | Type : COMPREHENSION

Given the vector $\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and the line passing through the origin represented

by the vector $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Answer the given subquestions.

Discussions (0)



Question 12 : 640653992014

View Parent QN

View Solutions (0)

Total Mark : 3.00 | Type : SA

Find the length of the projection of $\mathbf{x}$ onto the line defined by $\mathbf{w}$.

Enter your answer correct to two decimal places.

Answer (Numeric):

Answer

Discussions (0)



Question 13 : 640653992015

View Parent QN

View Solutions (0)

Total Mark : 4.00 | Type : SA

Calculate the magnitude(norm) of reconstruction error after projecting $\mathbf{x}$ onto the line defined by $\mathbf{w}$.

Enter your answer correct to two decimal places.

Answer (Numeric):

Answer

Solution(12)

$$\Rightarrow \text{formulae for projection} = \frac{(x^T w)w}{w^T w} = \frac{1}{2} \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \text{length of projection} = \left\| \begin{bmatrix} \frac{7}{2} \\ \frac{7}{2} \end{bmatrix} \right\| = \sqrt{\left(\frac{7}{2}\right)^2 + \left(\frac{7}{2}\right)^2} = \sqrt{12.25 + 12.25} \approx 4.95$$

$$\Rightarrow \text{Error vector} = x - \frac{(x^T w)w}{w^T w} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 7 \\ 7 \end{bmatrix} = \begin{bmatrix} \frac{-1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Solution(13)

$$\Rightarrow \text{norm of error vector} = \left\| \begin{bmatrix} \frac{-1}{2} \\ \frac{1}{2} \end{bmatrix} \right\| = \sqrt{\left(\frac{-1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{0.25 + 0.25} = \sqrt{0.5} \approx 0.71$$